

continuation

Continuation Document

Revision: 85 **Last modified:** 2026-06-01T23:30:49Z **Description:** Sacred invariant resumption document for Helix Cluster OS **Authority:** Constitution §12.10
Maintainer: Operator + AI loop

§0: How to use this document

Any CLI agent resuming work on this project MUST read this file first.

§1: Recently Completed Work (last 10 commits)

Commit	Message
aba565b	Foundation wave 25: 2 new test-orchestration pkgs + 1 verified-already-done. HXC-1254 parallel TestRunner (NEW pkg/testing/runner: Suite registration + bounded-worker-pool parallel execution + result collection/aggregate Report sorted deterministically; barrier-proof concurrency test (serial runner deadlocks under a ctx-deadline guard), atomic peak counter proves MaxConcurrency cap, result counts incl Err->Failed (defense-in-depth: both guards must be removed to break the want-3-got-4 bite — mutation-verified), Register dup/empty-name errors). HXC-1255 session test FSM (NEW pkg/testing/sessionfsm: Idle->Setup->Running->ChaosInject->Verify->Done + any-nonterminal->Failed; legal-transition table, IllegalTransitionError leaves state unchanged, per-state callbacks fire in order, History(); mutation guard-always-allow -> illegal transition test FAILS). HXC-1267 helix-test CLI VERIFIED already complete (cmd/helix-test dst/chaos/device/snapshot subcommands wired to real pkg/testing packages w/ 13 mutation-sensitive tests; device subcmd lists real T1-T8) — verified+marked, not rebuilt. ZERO new deps, ZERO edits to existing files. Gate: build/vet/vet-integration clean, full -short -race green, dataplane ok, tmutex clean (W23 fix holds); mutation spot-checks: FSM guard bites, TestRunner Err->Failed bites (both guards), cap-peak<=2 + barrier verified by review.
ace79d4	Foundation wave 24: mark HXC-1162 Completed — E2E multi-node session CRDT convergence (5 replicas, arbitrary-order gossip, converge to identical Fingerprint + canonical truth; mutation-verified bite). Registry 187->188 Completed / 451 Queued.
8e8cebb	Foundation wave 24: HXC-1162 E2E multi-node session CRDT convergence (pkg/session/convergence_hxc1162_test.go, test-only, ZERO edits). Proves

Commit	Message
	STRONG EVENTUAL CONSISTENCY beyond the existing 2-replica tests: 5 independent replicas each seeded with a DISJOINT partition of 100 ops, gossiped via seeded-random ARBITRARY-ORDER pairwise merges (<code>dst = dst.Clone().Merge(src)</code>) until 3-sweep quiescence; asserts (a) ALL 5 replicas reach an identical Fingerprint AND (b) that fingerprint equals a canonical truth replica that applied all 100 ops sequentially — over 20 seeds. Reuses existing <code>genUpdates/replicaFrom/buildUpdate + CRDTSessionState Merge/Clone/ Fingerprint</code>. Builds on W22 <code>pkg/checkpoint_merge</code>: this is the convergence guarantee that makes coordinator-free session migration safe. Gate: <code>build/vet/ vet-integration clean, full -short -race green</code> (zero failures this run), <code>dataplane ok, tmux ls clean</code> (W23 leak fix holds); INDEPENDENT MUTATION SPOT-CHECK: order-dependent equal-version pane tiebreak (<code>paneWins 'return false'</code>) -> replicas DIVERGE -> 'CONVERGENCE FAILED replica 0 and 1 diverged' (the N-node convergence bite is genuinely real). ZERO new deps. Foundation wave 23: fix <code>tmux-session</code> test leak (CLAUDE-1/§7.1 test hygiene; PTY-exhaustion guard, W15/W22 lesson). Real-<code>tmux</code>-backed tests now leave ZERO leaked sessions. ROOT CAUSE was <code>cmd/helix-session/main_test.go: TestRunServesRealRPC + TestRunGracefulShutdownStopsServing</code> call the <code>production run()</code> (real <code>tmux</code> backend since W15) and <code>CreateSession 2</code> real <code>tmux</code> sessions (<code>server-assigned session- id == tmux session name</code>) with no cleanup -> leaked on every full-suite run. FIX: added <code>killSession(t,id)</code> helper <code>t.Cleanup-ing 'tmux kill-session -t '</code> after each <code>CreateSession</code>. Also <code>pkg/session/backends/ package_test.go</code> (W23 workflow): added <code>t.Cleanup</code> kills to <code>TestTmuxBackend_CreateAndKill/Lifecycle</code> (defended against mid-test-failure leaks). VERIFIED: <code>full -short -race ./...</code> then '<code>tmux ls</code>' => no server running (zero leaks); real-<code>tmux</code> proofs intact (<code>hxc1136</code> exit gate untouched). Test-only changes, ZERO production code modified. NOTE flakes: <code>internal/advisory TestLockBlocksUntilAvailable + test/e2e TestClusterLifecycleSuite</code> FAIL only under heavy concurrent full-suite -race load, PASS isolated (load-induced timing flakes, not regressions).
06ca1a2	
639e479	Foundation wave 22: mark 2 items Completed (HXC-1152/1160) — container-checkpoint migration CRDT merge / <code>tmux</code> full functional Attach (<code>send-keys+capture-pane</code>, real marker round-trip). Mutation spot-checks confirm bites. Registry 185->187 Completed / 452 Queued.
1d79b5c	Foundation wave 22: 2 disjoint streams (both approved no-fix; mutation spot-checks bite). HXC-1152 container-checkpoint migration CRDT merge (NEW <code>pkg/checkpoint_merge: Apply(local,checkpoint)</code> restores+merges incoming CRDT checkpoint into a clone of local via existing <code>CRDTSessionState.Merge LWW, Migrate(source,local)</code> models <code>export-A->merge-B</code>; NEW <code>pkg/session/ crdt_checkpoint.go v2</code> checkpoint helper avoids import cycle; convergence tests assert EXACT LWW winner values + commutativity-by-Fingerprint + idempotence + corrupt-checkpoint sentinel; mutation skip-merge -> LWW winner reverts to wrong value FAILS). HXC-1160 <code>tmux</code> full functional Attach (<code>pkg/session/backends/tmux_attach.go + tmux.go Attach</code>: replaced the dead orphaned-<code>io.Pipe</code> stream with a REAL stream — <code>Write</code> delivers keystrokes via <code>tmux send-keys -l/Enter</code>, <code>Read</code> returns REAL pane content via <code>tmux capture-pane -p</code>; marker round-trip test against real <code>tmux</code> with bounded 20x100ms poll, unique names + defer Kill, ZERO leaks; mutation neuter <code>send-keys -> marker absent</code> FAILS). ZERO new deps. Gate: <code>build/vet/vet-integration clean, full -short -race green, dataplane ok</code>; both mutation spot-checks bite; W22 <code>tmux</code> test verified zero-leak in isolation. NOTE pre-existing leak: <code>pkg/session.Manager-</code>

Commit	Message
	based tests (manager.go:106 session- naming, e.g. hxc1136) can leave tmux sessions — hygiene item, not introduced here. Foundation wave 21: mark 3 items Completed (HXC-1130/1249/1243) — screen+zellij session backends (real screen, zellij ErrUnsupported) / emergency-stop+auto-recovery / 8 network fault injectors (verified already complete). Mutation spot-check confirms emergency-stop bite. Registry 182->185 Completed / 454 Queued. Foundation wave 21: 2 impl streams + 1 verified-already-done. HXC-1130 screen+zellij session backends (pkg/session/backends/screen.go real exec of /usr/bin/screen: Create -dmS / List -ls parse / Kill -X quit / SendInput -X stuff / CaptureOutput -X hardcopy; REAL integration test creates+lists(sink-side proof)+kills a uniquely-named session w/ defer-Kill, ZERO leaked sessions verified; zellij.go returns typed ErrBackendUnavailable since zellij absent — no fake; orchestrator fixed the Attach stream to deliver Write as REAL keystrokes via SendInput instead of an orphaned io.Pipe). HXC-1249 emergency-stop + auto-recovery (pkg/testing/scenario/stop.go: StopController.EmergencyStop snapshots ActiveFaults + RestoreAll ->
09d0044	StopResult{FaultsRecovered,HaltLatencyMS,WithinBudget,RecoveredNames}; halt latency modelled as data (faults*perFaultCost), NO real sleep; AutoRecover asserts no faults remain; post-stop ActiveFaults()==0 + WithinBudget false-case are the bites). HXC-1243 8 network fault injectors VERIFIED already complete (NetworkPartition/PacketLoss/LatencyInjection + PacketReorder/PacketDuplication/BandwidthThrottle/AsymmetricPartition/DNSBlackhole, all with passing tests) — verified+marked, not rebuilt. ZERO new deps. Gate: build/vet/vet-integration clean, full -short -race green, dataplane ok; mutation spot-check: B skip-RestoreAll -> post-stop ActiveFaults!=0 FAILS (bite real); A real screen create/list/kill verified sink-side, zero leaks.
15977c0	Foundation wave 20: mark 2 items Completed (HXC-1248/1244) — YAML chaos scenario engine (blast-radius+abort) / simulated node fault injectors. Plus HXC-1110 /metrics coverage completed for the 6 gRPC-only services. Mutation spot-checks confirm bites. Registry 180->182 Completed / 457 Queued. Foundation wave 20: 3 disjoint streams (all approved; A minor silent-error-discard fixed by orchestrator). /metrics sidecar completing HXC-1110 ‘every binary’ intent (pkg/metrics/sidecar.go StartSidecar/StartSidecarFromEnv opt-in HTTP listener HELIX_METRICS_ADDR-gated; in-process scrape asserts exact ‘helix_sidecar_hits_total 3’; wired into the 6 gRPC-only mains advisory/build/node/scheduler/security/session; orchestrator added log-on-bind-error to all 6 to avoid silent failure). HXC-1248 YAML chaos Scenario Engine (NEW pkg/testing/scenario: Scenario{phases,faults,MaxFaults,MaxAffected,AbortOn} parsed via yaml.v3; Engine.Run->ExecutionLog with per-phase + global blast-radius clamp + abort-on-condition early-stop; deterministic, no wall-clock; reuses chaos catalog read-only). HXC-1244 simulated node fault injectors (NEW pkg/testing/chaos/node_faults.go: NodeRestart w/ recovery window, DiskSpaceExhaustion write-block, ProcessTerminate — each a real chaos.Fault with state-flipping Apply/Restore; distinct from existing NodeCrash/SlowDisk/OOMKill). ZERO new deps. Gate: build/vet/vet-integration clean, full -short -race green, dataplane ok; mutation spot-checks B blast-radius + C node-restart bite; A sidecar exact-value scrape verified.
bc99718	
e93fad6	

§2: Environment Snapshot

Property	Value
Branch	main
Commit	aba565b
Timestamp	2026-06-01T23:30:49Z

§3: Active Work

HXC	Title	Status
HXC-922	Node service, advisory locks, chaos tests, docs	✓ Done
HXC-921	Web UI, K8s/Helm, integration, E2E, benchmarks	✓ Done
HXC-920	Health, security, build, wireguard services	✓ Done
HXC-919	htmux, messaging, GPU, WASM, storage, build manifest	✓ Done
HXC-918	Wire scheduler/session gRPC to real backends, etcd discovery	✓ Done
HXC-916	Stub expansion (config, retry, ratelimit, validator, build cache)	✓ Done
HXC-914	LLM, policy, setup, distributed lock	✓ Done
HXC-913	Health, security, ClassAds parser	✓ Done
HXC-912	Gateway, helixd, helix-agent	✓ Done
HXC-911	Scheduler, Session gRPC services	✓ Done
HXC-910	Console, security, testing infra	✓ Done

§4: Next Planned Work

All MVP, Phase 2, Phase 3, and Phase 4 are **COMPLETE**. Potential future enhancements:

1. **Performance optimization** — Profile and optimize hot paths
2. **Multi-region support** — Cross-region cluster federation
3. **GPU scheduling v2** — Multi-GPU, fractional GPU allocation

4. **Web UI real-time** — WebSocket integration for live updates
5. **Operator pattern** — Kubernetes operator for cluster management

§5: Known Issues / Blockers

- None

§6: Quick Commands

Run all tests (requires GOWORK due to workspace modules)

```
GOWORK=$(pwd)/go.work go test ./... -race -count=1
```

Build all binaries

```
GOWORK=$(pwd)/go.work go build ./cmd/...
```

Build web UI

```
cd web && npm run build
```

Render Helm chart

```
helm template helix-cluster deploy/helm/
```

Run benchmarks

```
go test -bench=. ./test/benchmark
```

Run chaos tests

```
go test -race -count=1 ./test/chaos
```